

Condition-Based Maintenance & Vibration Monitoring SOP

Plant standard operating procedure for online CBM

Field	Value
Document No.	SOP-CBM-0007
Revision	Rev 3.1
Equipment OEM	(synthetic OEM)
Applicable standard	ISO 10816 / IEC / CEA norms
Classification	O&M reference - SYNTHETIC (Ballast demo)
Owner	Plant O&M / Reliability

Disclaimer: This is a synthetic document generated for the Ballast demo. Values are grounded in public standards and typical industry practice but do not represent any specific OEM's proprietary manual.

1. Purpose & Scope

Defines vibration/temperature monitoring, alarm philosophy and the escalation workflow for rotating equipment (mills, fans, BFPs, CW/CE pumps, turbine-generator). Aligns with ISO 10816 severity zones and the plant's predictive-maintenance strategy.

2. ISO 10816 Severity Zones (velocity RMS, mm/s)

Zone	Support A (rigid)	Support B (flexible)	Meaning
Zone A (new/good)	≤ 2.3	≤ 3.5	Newly commissioned; ideal
Zone B (acceptable)	2.3 - 4.5	3.5 - 7.1	Unrestricted long-term operation
Zone C (alarm)	4.5 - 7.1	7.1 - 11.0	Short-term only; plan corrective action
Zone D (danger)	> 7.1	> 11.0	Risk of damage; trip / immediate action

3. Measurement & Alarm Philosophy

- Measure at each bearing housing in H, V, A directions; trend overall RMS + spectra.
- ALERT at Zone C entry (typ. 4.5 mm/s): raise CBM notification, tighten monitoring.
- DANGER at Zone D entry (typ. 7.1 mm/s): ready standby, prepare controlled derate.
- TRIP on sustained Zone D per equipment set-point.
- Trend matters more than absolute value: a rising slope over days warrants action even within Zone B.

4. Escalation Workflow

Step	Trigger	Action	Owner
1	Zone C entry	Auto CBM alert + predictive WO raised	CBM system
2	Trend confirmed	Diagnose (spectra, oil analysis)	Reliability engineer
3	No healthy standby / spare	Assess commercial risk; plan derate	Shift + Commercial
4	Zone D / imminent	Controlled derate or transfer; execute WO	Shift charge engineer

Commercial linkage: before any forced trip, evaluate PAF headroom vs NAPAF and DSM exposure. A pre-emptive controlled derate is preferred to an uncontrolled trip (see ABT primer).